Dental Caries

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Introduction

Dental caries or decay, an almost ubiquitous clinical problem in children has been reported in nearly 30–60% of preschool children across the globe, including India. Benjamin RM reported that it is one of the most common chronic diseases in children being five times more common than asthma or hay fever.

Definition

Dental caries is the localized destruction of susceptible dental hard tissues by acidic by-products from bacterial fermentation of dietary carbohydrates.

Etiology

- ☑ Multifactorial disease
- ☑ Three main factors are responsible, i.e., oral bacteria in dental plaque, presence of fermentable carbohydrates, and available tooth surface.
- ☑ Contributing factors that increase or decrease the speed of decay are oral hygiene habits, shape of the tooth, surface characteristics, eating habits, and quality and quantity of saliva.
- ☑ *Other factors:* Poverty, deprivation, or social status; number of years in education; dental insurance coverage; use of dental sealants; use of orthodontic appliances; and poorly designed or ill-fitting partial dentures.
- ☑ *In older children:* Alignment and spacing of the early permanent dentition, dietary changes, and exposure to orthodontic appliances.

Pathogenesis

- ☑ Bacteria (*Streptococcus mutans, Streptococcus sobrinus,* and *Lactobacillus* species) in the biofilm produce weak organic acids as a by product of metabolism of fermentable carbohydrates.
- ☑ This acid causes local pH values to fall below a critical value resulting in demineralization of tooth tissues. If the diffusion of calcium, phosphate, and carbonate out of the tooth is allowed to continue, cavitation will eventually take place.

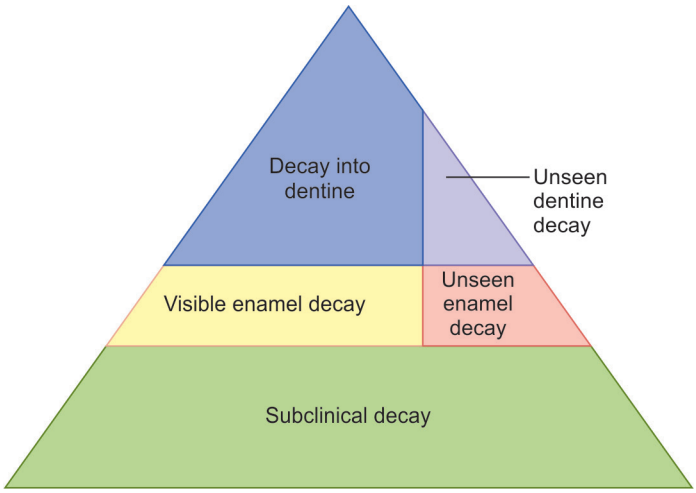

Fig. 1: Stages of tooth decay.

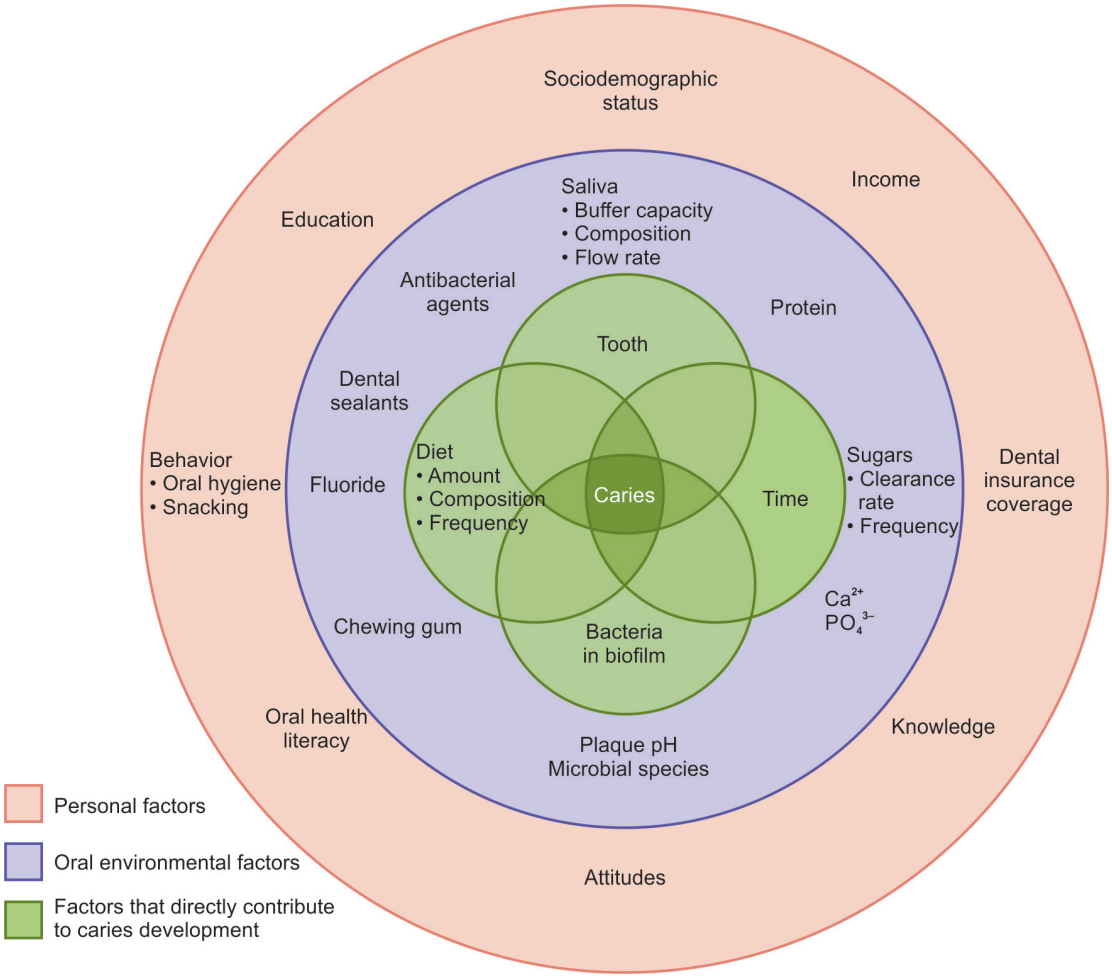

Fig. 2: Causative factors of dental caries.

The most commonly used diagnostic tool is visual inspection along with tactile (probing) examination. The teeth should be cleaned, dried, and examined in good lighting. A whitish carious spot is detected more easily when the tooth is dry. Caries is initially recognized as a dull, white band of demineralized enamel which on further follow-up quickly advances to obvious decay along the gingival margin. The decayed hard tissue is evident as a yellow or brown cavitated area. It is generally first seen on the primary maxillary incisors, and all the four maxillary anterior teeth are usually involved concurrently. In permanent teeth, the first and second molars are the most caries prone. The lesions may be found on either the labial or lingual surfaces of the teeth and, in some cases, on both. The occlusal or biting surfaces of teeth account for most of the caries.

Occasionally, it can present as an unexplained fever resulting from an abscess in the jawbone associated with a carious tooth, or as submandibular or deep cervical lymphadenopathy due to spread of infection to the root of the tooth.

The DMFT index is the sum of the number of decayed, missing due to caries, and filled teeth in the permanent teeth. An index ≥4 at the age of 3 years, 5 at the age of 4 years, and 6 at the age of 5 years denotes severe early childhood caries.

The International Caries Detection and Assessment System (ICDAS) used to grade the severity of caries is as follows:

- ☑ **Sound surface (Code-0)**
- ☑ **First visual change in enamel (seen only after prolonged air-drying or restricted to within the confines of a pit or fissure) (Code-1)**
- ☑ **Distinct visual change in enamel (Code-2)**
- ☑ **Localized enamel breakdown (without clinical visual signs of dentin involvement) (Code-3)**
- ☑ **Underlying dark shadow from dentin (Code-4)**
- ☑ **Distinct cavity with visible dentin (Code-5)**
- ☑ **Extensive distinct cavity with visible dentin (Code-6)**

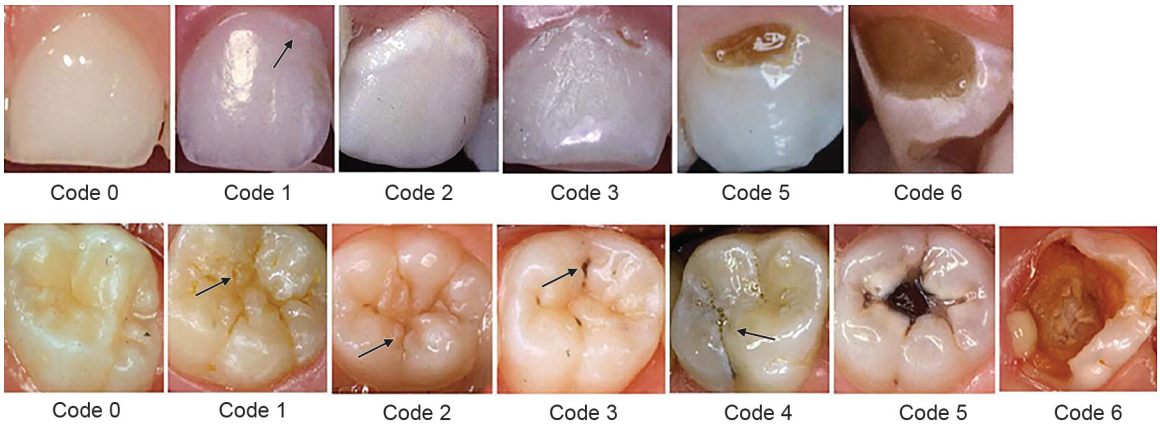

Fig. 3: International caries detection and assessment staging-various codes.

In addition, digital panoramic X-rays (Panorex) and periapical dental films are useful to identify involvement of tooth and surrounding bone. Blood investigations like complete blood count (CBC) are usually noncontributory.

Treatment

The focus is on prevention of dental caries and on preventing progression of early cases, so that the need for surgical/restorative intervention is minimized.

So now the strategies are:

- ☑ Active surveillance (prevention therapies and close monitoring) of enamel lesions
- ☑ Use of fluoridated toothpaste
- ☑ Radiographic diagnostic recommendations
- ☑ Pit and fissure sealants
- ☑ Dietary interventions
- ☑ Caries risk assessment at both the individual level and at the tooth level
- ☑ Treatment of caries with interim therapeutic restorations

Topical silver diamine fluoride, a topical medication in which the silver acts as an anti-microbial and the fluoride promotes the remineralization of the tooth is considered to be an effective local therapy. Recommended maximum dose is one drop/25 µL per 10 kg per weekly treatment. The undesirable side effect is darkening of the carious lesions.

Systemic amoxicillin is the first-line drug of choice to treat any infection and prevent further complications, but with increasing resistance, alternatives of amoxicillin/clavulanate, clindamycin, and metronidazole are becoming preferred choice of clinicians.

Nonoperative treatment aims at reversing, arresting, or postponing progression of the caries lesion and should be the choice of treatment whenever possible. Beyond that when surgical intervention or restorative procedures are required; the children should be referred to qualified pediatric dentists.

Prevention

Pediatricians should identify *risk factors that increase the risk of dental caries*: Breastfed/bottle fed beyond 1 year of age, night-time bottle feeds, sugary beverages and snacks, prolonged use of sippy cup, special-need child, caregiver with caries, low socioeconomic status, passive tobacco smoke, and visible enamel pits or defects.

Role of Pediatrician

Ideally, a dental referral should be made for all children at 1 year of age. However, this may not be practical. So a pediatrician should perform dental risk evaluation (caries and malocclusion), and ensure early referral of children at high risk and those in whom early signs are detected to a dentist. They should also offer advice on dental hygiene, diet, dietary habits, and oral habits.

Prevention

Anticipatory Guidance

Brushing should be started with the eruption of the first tooth, with a soft small toothbrush, twice a day, without toothpaste. Starting early helps to minimize "refusal to brush". For children, low fluorine toothpastes are available. These can be started from the second year, using a "grain" size for <3 years and "pea" size toothpaste for 3–6 year-old children. Pediatric toothpastes foam less so ingestion and subsequent side effects are lesser. Children >6 years can use a bit more than pea size fluoridated toothpaste.

Preventive Advice

Besides addressing the preventable risk factors from those enumerated above, parents should be advised to follow the following: combine a sugary snack with protein and/or fat or with the meal, sugary drink to be consumed in limited time and not sipping throughout the day, avoid grazing through the day, rinse mouth with water immediately after consumption, and if on bottle feeding, to discontinue bottle as early as possible, latest by 1 year of age.

Further Reading

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